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### TRANSIENT DYNAMICS OF THREE-DIMENSIONAL BEAM TRUSSES USING HIGHER ORDER KINEMATICS

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#### ABSTRACT

Spatial structures may be subjected to impulse loads which give rise to the propagation of high-frequency (HF), strongly oscillating waves. Despite some recent researches, the characterization of the transient response of engineering systems to such loads remains an open problem. The objective of this research is to develop a reliable model of the HF energy evolution within three-dimensional beam trusses in order to predict, for example, their potential steady-state behavior at late times or the energy paths. The theory of micro-local analysis of linear wave systems shows that the energy density associated with their HF solutions satisfies a Liouville-type transport equation. A suitable transport model for beams is derived from Timoshenko kinematics, and subsequently illustrated by the dispersion relations for HF Rayleigh-Lamb waves in a waveguide. At the interfaces between substructures, the energy flow is partly reflected and partly transmitted. The corresponding power flow reflection/transmission coefficients have also been derived. Numerical simulations are performed by nodal or spectral discontinuous Galerkin (DG) methods for spatial discretization and a strong stability-preserving Runge-Kutta (RK) method for time integration. Numerical results using the RK-DG method are presented for the example of a three-dimensional beam truss that exhibits a diffusive behavior at late times.

#### INTRODUCTION

High-frequency (HF) wave propagation may be generated within three-dimensional beam trusses by impulse or shock loads. In this range, classical finite element methods are not applicable on account of the small wavelengths and high modal densities. Therefore several heuristic strategies have been developed by engineers. Global approaches such as the Statistical Energy Analysis (SEA) [1] of structural-acoustic systems give information on the mean energy level in each substructures. The main issue with SEA lies in the estimation of some core parameters such as the coupling loss factors or the power inputs. Local approaches such as the Vibrational Conductivity Analogy (VCA) [2] have been also investigated. The point of issue concerning the existing local approaches is their extension to complex structures. The approach adopted here is to consider an energy transport formulation for the energy density carried by HF waves.

Based on microlocal analysis of classical wave systems, it can be proved that the energy density associated with their HF solutions satisfies a Liouville-type transport equation. This theory can be adapted to quantum wave, classical acoustic or electromagnetic waves as shown by Papanicolaou & Ryzhik [3]. Transport equations can be solved numerically by various methods including Monte-Carlo methods [4] or nodal/spectral discontinuous Galerkin (DG) finite element methods [5]. Both techniques have the advantage to be weakly dissipative and weakly dispersive (depending on the order of interpolation). Thus they are well suited for late-time simulations in order to exhibit the diffusive behavior of the transport equations [6]. These methods have been

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applied to assemblies of two-dimensional beams or plates [7], as well as to randomly heterogeneous shells [8,9].

The main objectives of this paper are threefold. First, the two-dimensional model of HF energy evolution derived by Savin [7] is extended to a curved, three-dimensional beam model for application to multi-bay trusses. Second, a general framework for numerical simulations by discontinuous finite elements is derived. This mathematical-numerical model of the transient dynamics of beam trusses shall be able to predict, for example, their diffusive behavior at late times or the energy paths within the different bays. Third, the relevance of choosing Timoshenko kinematics for the description of HF wave propagation is discussed. In the first part of this paper (next section), the mechanical model of a curved, three-dimensional Timoshenko beam is recalled and his transient HF behavior is derived from the aforementioned transport theory. In order to extend this theory to HF wave propagation in multi-bay trusses, the various reflection/transmission phenomena at junctions of beams are also briefly analyzed in this section. The resulting coupled transport equations are numerically integrated by a discontinuous finite element method. In the second part of the paper, the validity of Timoshenko kinematics is discussed and compared to an asymptotic analysis of the Rayleigh-Lamb frequency spectrum of circular waveguides. The last section of the paper draws some conclusions.

## HF DYNAMICS OF BEAM TRUSSES USING TIMOSHENKO KINEMATICS

In this section, both the derivation and the implementation of the transport model for beam networks are presented. An illustrative example allows to highlight some properties of the HF transient response of such structures.

### HF Behavior of Timoshenko Beam Trusses

The application of the transport theory first requires to establish the dynamic behavior of a three-dimensional Timoshenko beam, *i.e.* its dynamic and constitutive equations. The purpose of this section is to summarize these equations and to present a modeling of HF wave propagation in beam assemblies considering reflection/transmission phenomena at the junctions. More details may be found in [10].

**Geometry and Kinematic Hypotheses.** The vibrational behavior of the beam is modeled by Timoshenko's beam kinematics. It includes the effects of rotary inertia and shear distortion [11]. Another important point in this theory is the introduction of a reduced shear modulus to account for the variations of the shear stress along the cross-section. The latter is assumed to be geometrically symmetric such that its geometrical center is on the neutral fiber of the beam. In this paper beam's material

is assumed to be isotropic and its heterogeneities vary slowly in regard to the considered wavelengths. The model below allows for curvature and torsion of the neutral fiber. Let  $\hat{\mathbf{t}}(s)$ ,  $\hat{\mathbf{n}}(s)$ ,  $\hat{\mathbf{b}}(s)$  be the tangent, normal and bi-normal unit vectors constituting a Frenet-Serret frame defined about the neutral fiber  $\mathcal{C}$ , which is parametrized by the curvilinear abscissa  $s$  in a subset  $\mathcal{S}$  of  $\mathbb{R}$ . The corresponding curvilinear coordinates are  $\mathbf{s} = (s, s_2, s_3)^T$  such that a point  $\mathbf{x}$  in the subset  $\Omega$  of  $\mathbb{R}^3$  occupied by the beam is parametrized by  $\mathbf{x} = \Psi(\mathbf{s})$  in a fixed reference frame.  $\Omega$  is also written  $\Omega \equiv \mathcal{C} \times \Sigma(s)$ , where  $\Sigma$  is the beam cross-section. Then  $\hat{\mathbf{t}}(s) = d\mathbf{x}/ds$ ,  $\hat{\mathbf{n}}(s) = |d\hat{\mathbf{t}}/ds|^{-1} d\hat{\mathbf{t}}/ds$ , and  $\hat{\mathbf{b}}(s) = \hat{\mathbf{t}} \times \hat{\mathbf{n}}$ . The curvature  $\kappa(s)$  is defined by  $d\hat{\mathbf{t}}/ds = \kappa\hat{\mathbf{n}}$ , and the torsion  $\tau(s)$  is defined by  $d\hat{\mathbf{b}}/ds = -\tau\hat{\mathbf{n}}$ . The cross-section dimensions have to be small in regard to both the curvature and the torsion [12]. By Timoshenko's beam kinematical assumption, the displacement of a point  $\mathbf{s}$  of the cross-section is written in the local frame:

$$\mathbf{U}(\mathbf{s}, t) = \mathbf{U}_c(s, t) + \boldsymbol{\Theta}(s, t) \times \mathbf{s}_\perp, \quad (1)$$

where  $\mathbf{s}_\perp = (0, s_2, s_3)^T$ ,  $\mathbf{U}_c(s, t) = (u_c(s, t), v_c(s, t), w_c(s, t))^T$  is the displacement of the neutral fiber in the local Frenet frame, and  $\boldsymbol{\Theta}(s, t) = (\theta_1(s, t), \theta_2(s, t), \theta_3(s, t))^T$  is the small rotation vector of the cross-section in this very frame.

### Balance of Momentum and Constitutive Equations.

Under the assumption of small perturbations, the dynamical equilibrium and the constitutive equations of the rigid cross-section at  $s$  within a three-dimensional curved beam, integrated over the cross-section, read:

$$D\mathbf{F} = \rho S \partial_t^2 \mathbf{U}_c, \quad (2)$$

$$D\mathbf{M} + \hat{\mathbf{t}} \times \mathbf{F} = J \partial_t^2 \boldsymbol{\Theta}, \quad (3)$$

$$\mathbf{F} = S \mathbf{C}_1 (D\mathbf{U}_c + \hat{\mathbf{t}} \times \boldsymbol{\Theta}), \quad (4)$$

$$\mathbf{M} = \mathbf{C}_2 D\boldsymbol{\Theta}, \quad (5)$$

where  $\rho(s)$  is the material volume density and  $S$  the cross-section area;  $\mathbf{F}(s, t) = (T_1(s, t), T_2(s, t), T_3(s, t))^T$  and  $\mathbf{M}(s, t) = (M_1(s, t), M_2(s, t), M_3(s, t))^T$  are respectively the resultant forces and resultant moments acting on the cross-section in the local frame;  $\mathbf{J}(s) = \text{diag}(I(s), J(s), K(s))$ ,  $J = \rho \int_\Sigma s_3^2 d\Sigma$ ,  $K = \rho \int_\Sigma s_2^2 d\Sigma$ ,  $I = J + K$  are the inertias of the beam;  $\mathbf{C}_1(s) = \text{diag}(E, \mu', \mu')$  is the relaxation tensor for translational motion, with  $\mu'(s)$  the reduced shear modulus averaged over the cross-section, and  $E(s)$  is the Young modulus averaged over the cross-section;  $\mathbf{C}_2(s) = \text{diag}(\mu' I, EJ, EK)$  is the average relaxation tensor for rotational motions; and  $D(s)$  is the so-called Frenet matrix induced by the derivation with respect to the curvilinear abscissa

along the neutral fiber:

$$\mathbf{D}(s) = \begin{bmatrix} \partial_s - \kappa & 0 \\ \kappa & \partial_s - \tau \\ 0 & \tau & \partial_s \end{bmatrix}. \quad (6)$$

More details on the derivation can be found in [10, 13]. Now introducing the state vector  $\mathbf{X}(s, t) = (\partial_t \mathbf{U}_c^T, \partial_t \boldsymbol{\Theta}^T, \mathbf{F}^T, \mathbf{M}^T)^T$  in  $\mathbb{R}^{12}$ , Eq. (2) and Eq. (5) have the form of an  $12 \times 12$  first-order system:

$$\begin{cases} \mathbf{A}(s) \partial_t \mathbf{X} = \mathbf{P}(s, \partial_s) \mathbf{X}, \\ \mathbf{X}(0, s) = \mathbf{X}_0(s), \end{cases} \quad (7)$$

where  $\mathbf{A}$  is the block diagonal matrix defined by  $\mathbf{A}(s) = \text{diag}(\rho S \mathbf{I}_3, \mathbf{J}, (S \mathbf{C}_1)^{-1}, \mathbf{C}_2^{-1})$ ,  $\mathbf{I}_n$  is the  $n \times n$  identity matrix, and  $\mathbf{P}(s, \partial_s)$  is the operator:

$$\mathbf{P}(s, \partial_s) = \begin{bmatrix} 0 & 0 & \mathbf{D} & 0 \\ 0 & 0 & \boldsymbol{\Omega} & \mathbf{D} \\ \mathbf{D} & \boldsymbol{\Omega} & 0 & 0 \\ 0 & \mathbf{D} & 0 & 0 \end{bmatrix}, \quad \boldsymbol{\Omega} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad (8)$$

which may also be written  $\mathbf{P}(s, \partial_s) = \mathbf{D}^1 \partial_s + \mathbf{D}^0(s)$ ;  $\mathbf{D}^1$  and  $\mathbf{D}^0$  are  $12 \times 12$  matrices obtained straightforwardly from Eq. (8) above and Eq. (6). Note that  $\mathbf{D}^1$  is symmetric and has elements equal to 0 or 1, and  $\mathbf{D}^0$  is skew-symmetric. At last  $\mathbf{X}_0(s)$  is a vector of initial conditions.

**HF Transport Model for a Three-Dimensional Beam.** HF propagation corresponds to a highly oscillatory wave in space, that is very difficult to describe by the classical displacement wave equation. Contrarily, the energy wave is much smoother and shall be the quantity of interest for our model. In this context, it is possible to show that the HF energy density for a given cross-section abscissa  $s$  is expressed as:

$$\mathcal{E}(s, t) = \sum_{\alpha \in E} \int_{\mathbb{R}} w_{\alpha}(s, k, t) dk, \quad (9)$$

where  $E = \{P_i, T_i, 1 \leq i \leq 3\}$  are the indices of the energy modes.  $P_i$  denotes the compressional modes gathering the longitudinal and the two bending modes, while  $T_i$  denotes the transverse modes gathering the torsional and the two shear modes. Their velocities are respectively  $c_P = \sqrt{E/\rho}$  and  $c_T = \sqrt{\mu'/\rho}$ . The power flow density vector  $\boldsymbol{\Pi} = \Pi \hat{\mathbf{k}}$  is also useful to investigate on the propagation of the energy density. In the HF range, it

is given by:

$$\boldsymbol{\Pi}(s, t) = \sum_{\alpha \in E} \int_{\mathbb{R}} \boldsymbol{\pi}_{\alpha}(s, \mathbf{k}, t) d\mathbf{k}, \quad (10)$$

where  $\boldsymbol{\pi}_{\alpha}(s, \mathbf{k}, t) = c_{\alpha}(s) w_{\alpha}(s, k, t) \hat{\mathbf{k}}$ , and  $\hat{\mathbf{k}}$  is the direction of propagation *i.e.* the sign of the wavenumber,  $\hat{\mathbf{k}} := \text{sign}(k) \hat{\mathbf{t}}$ . Finally it is possible to show that the evolution of the so-called specific intensities  $w_{\alpha}$  is described by the following transport equations:

$$\partial_t w_{\alpha} + c_{\alpha} \hat{\mathbf{k}} \partial_s w_{\alpha} - |k| \partial_s c_{\alpha} \partial_k w_{\alpha} = 0, \quad \alpha \in E. \quad (11)$$

**Power Flow Reflection/Transmission.** The HF energy density propagation in a beam is thus characterized by the transport equations (11). In order to use this model for complex three-dimensional beam trusses, the power flow reflections/transmissions at the junctions have to be considered. They contribute to couple the energy densities  $w_{\alpha}$ . These phenomena are analyzed through the reflexion/transmission power flow coefficients of a junction  $\rho_{\alpha\beta}^{11}$  and  $\tau_{\alpha\beta}^{q1}$ , of an arbitrary number  $\mathcal{N}$  of beams. They are expressed as:

$$\rho_{\alpha\beta}^{11} = \frac{\langle \Pi_{\alpha}^R \rangle}{\langle \Pi_{\beta}^I \rangle}, \quad (12)$$

$$\tau_{\alpha\beta}^{q1} = \frac{\langle \Pi_{\alpha}^{Tq} \rangle}{\langle \Pi_{\beta}^I \rangle}, \quad (13)$$

where  $\langle \Pi_{\alpha}^R \rangle$  is the time average power flow for the reflected waves in the beam #1 corresponding to the mode  $\alpha \in E$ ,  $\langle \Pi_{\alpha}^{Tq} \rangle$  is the time average power flow for the transmitted waves in the beam # $q$  corresponding to the mode  $\alpha$  in that beam, and  $\langle \Pi_{\beta}^I \rangle$  is the time average power flow for an incident wave in beam #1 in the mode  $\beta$ . The time average power flow are computed from the continuity of the displacement, rotation, forces and moments [7, 10]. It is observed from these results that in the HF limit, the reflection/transmission coefficients corresponding to translation incident motions  $u_c$ ,  $v_c$  or  $w_c$ , and rotational reflected/transmitted motions  $\theta_i$ ,  $i = 1, 2$  or  $3$ , vanish, as well as those related to rotational incident motions and translational reflected/transmitted motions. All other coefficients are independent of the wavenumber  $k$ . It means that, at higher frequencies, no rotational (respectively translational) motion arises when a translational (respectively rotational) wave impinges the junction.

### Numerical Simulation for a Truss

At junctions, power flows are partly reflected and partly transmitted, resulting in discontinuities of the energy density

field. Thus classical finite element method can not be used to solve numerically the Liouville equations (11). Indeed, although the overall power flow density is continuous across the junction owing to the Rankine-Hugoniot condition, the energy density itself does not necessarily have to be continuous across that junction. Therefore a discontinuous "Galerkin" (DG) finite element method [5, 14, 15] is used to solve Eq. (11) together with the reflection/transmission conditions. The theoretical background of the DG method is summarized below, more details may be found again in [10]. An illustrative example is presented in a subsequent paragraph.

**Discontinuous "Galerkin" Finite Elements.** The DG method for the transport equations (11) is written first on a single beam of neutral fiber occupying the bounded domain  $\mathcal{S}$ . Let us consider the subdivision  $\mathcal{T}_h = \cup_{r=1}^K D_r$  of  $\mathcal{S}$  into  $K$  non-overlapping sub-domains, or finite elements  $D_r$ . These elements are written  $D_r = [s_{r-\frac{1}{2}}, s_{r+\frac{1}{2}}]$  and the largest element size of this mesh is  $h = \max_{1 \leq r \leq K} |s_{r+\frac{1}{2}} - s_{r-\frac{1}{2}}|$ . It is assumed that the velocities  $c_\alpha$  are constant over the elements; they are denoted by  $c_\alpha^r$ . Here  $|k|$  is a fixed parameter, and the  $w_\alpha$ 's are considered as functions of  $\hat{k}$  rather than  $k$ . Moreover a globally defined space  $V_h$  is introduced as  $V_h = \oplus_{r=1}^K V_h^r$  where the locally defined test spaces  $V_h^r \subseteq H^1(D_r)$  do not enforce any particular boundary conditions on  $\partial D_r$ . It is thus possible to introduced a numerical flux  $\pi_\alpha^*$  which depends on the traces of  $w_\alpha$  from both sides of an interface between elements. This leads to the variational formulation:

$$\int_{D_r} \left( \frac{\partial w_\alpha}{\partial t}(s, \hat{k}, t) v(s) - c_\alpha^r \hat{k} w_\alpha(s, \hat{k}, t) \frac{dv}{ds} \right) ds = - [\pi_\alpha^* \cdot \hat{\mathbf{t}}^r v]_{s_{r-\frac{1}{2}}}^{s_{r+\frac{1}{2}}}, \quad \forall v \in V_h. \quad (14)$$

The numerical flux  $\pi_\alpha^*$  at the interface  $s = s_{r+\frac{1}{2}}$  is constructed from the computed values of  $w_\alpha(s_{r+\frac{1}{2}}, \hat{k}, t)$  in  $V_h^r$  (the trace of  $w_\alpha$  at  $s_{r+\frac{1}{2}}$  in  $D_r$ ) and  $w_\alpha(s_{r+1-\frac{1}{2}}, \hat{k}, t)$  in  $V_h^{r+1}$  (the trace of  $w_\alpha$  at  $s_{r+1-\frac{1}{2}}$  in  $D_{r+1}$ ) so as to be consistent and conservative, see [10] for more details.

Three main ingredients are needed to complete the numerical implementation of the DG method once the subdivision  $\mathcal{T}_h$  has been performed: (i) the definition of the numerical flux function  $\pi^*$  on all elements, (ii) the choice of the approximation set  $V_h$ , and (iii) the choice of a time-integration scheme for Eq. (14). A junction  $\{s_{r+\frac{1}{2}}\} = \{s_{r'-\frac{1}{2}}\} = \partial D_r \cap \partial D_{r'}$ ,  $r \neq r' \in \mathcal{J}_r$  of  $\mathcal{N}$  beams is considered.  $\mathcal{J}_r$  is the set of indices of the beam elements which are connected to the beam element  $\#r$ . The numerical fluxes ensuring the stability and consistency of the discontin-

uous Galerkin method for transport equations have the form:

$$\pi_\alpha^*(s_{r+\frac{1}{2}}, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^r = \sum_{\beta \in E} \left( \rho_{\alpha\beta}^{rr} \pi_\beta^r(s_{r+\frac{1}{2}}, \mathbf{k}_\beta^r, t) \cdot \hat{\mathbf{t}}^r + \sum_{r' \in \mathcal{J}_r} \tau_{\alpha\beta}^{rr'} \pi_\beta^{r'}(s_{r'-\frac{1}{2}}, \mathbf{k}_\beta^{r'}, t) \cdot \hat{\mathbf{t}}^{r'} \right) \quad (15)$$

for  $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^r < 0$ , and:

$$\pi_\alpha^*(s_{r+\frac{1}{2}}, t) \cdot \hat{\mathbf{t}}^r = \pi_\alpha^r(s, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^r \quad (16)$$

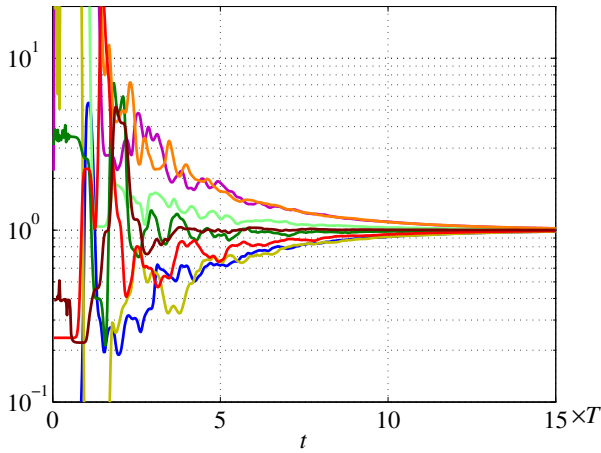
for  $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^r > 0$ . This choice corresponds to an upwind numerical flux. The approximation set  $V_h$  is constructed as follows:

$$V_h = \{v|_{D_r} \in \mathbb{P}_\ell(D_r), 1 \leq r \leq K\}, \quad (17)$$

where  $\mathbb{P}_\ell(D_r) \subset H^1(D_r)$  is typically a polynomial set. In our numerical simulations,  $\mathbb{P}_\ell(D_r)$  is spanned by a set of normalized Jacobi polynomials up to the order  $\ell$ . The numerical dispersion and dissipation errors introduced by this spatial discretization are  $O(K^{2\ell+3})$  and  $O(K^{2\ell+2})$ , respectively [16], where  $K = h|k|$ . These so-called "superconvergence" properties are very much desirable for long time simulations in order to exhibit the diffusion limit of the transport equations (11) with the reflection/transmission boundary conditions. Finally, time integration of the semi-discretized system (14) is performed by a strong stability-preserving high-order Runge-Kutta (RK) scheme [17].

**Numerical Example.** A beam truss constituted by  $\mathcal{N} = 8$  three-dimensional beams is considered: its shape is structured like a pyramid with a rectangular basis having side lengths of  $L = 5$  unit length ( $L_p = L$  for  $p = 1, 3$ ) on  $L = 4$  unit length ( $L_p = 4L$  for  $p = 2, 6$ ). The beam numbered  $p = 4$  is  $L = 5$  unit length long and the beam numbered  $p = 5$  is  $L = 3.6$  unit length long (see FIG. 1). The other geometrical quantities can be inferred from these informations. All beams have the same cross-sections of area  $S$  and are made from the same homogeneous, isotropic elastic material. Thus all beams have the same velocities  $c_p$  and  $c_T$  for the longitudinal-bending and transverse-torsional modes, respectively. The Poisson ratio is  $\nu = 0.3$  and the reduced shear modulus is  $\mu' = 5E/(12 + 10\nu)$ . The transport equations (11) hold in each beam and the reflection/transmission coefficients are directly computed from the approach outlined in the previous section. The energetic initial condition has a Gaussian shape and loads the mode  $P_1$  (a pure compressional wave). It is applied to, say, beam labelled #1 with a fixed initial direction  $\hat{\mathbf{k}} = +1$ . It is displayed with a red line on the FIG. 1. The nodal density is fixed at  $20/L$  uniform spatial elements per unit length





**FIGURE 3.** EVOLUTION OF THE RATIO  $\frac{c_T c_T^p(t)}{2 c_p c_p^p(t)}$  IN EACH BEAM.

vector field  $\nabla \times \Psi$  and a rotational-free vector field  $\nabla \Phi$  leads to:

$$\nabla^2 \Psi = \frac{1}{c_t^2} \partial_t^2 \Psi, \quad \Delta \Phi = \frac{1}{c_l^2} \partial_t^2 \Phi, \quad (19)$$

where  $c_l = \sqrt{(\lambda + 2\mu)/\rho}$  and  $c_t = \sqrt{\mu/\rho}$  are the phase velocity of longitudinal and transverse waves, respectively, for a three-dimensional medium. Assuming that the solutions of this problem are waves propagating in the positive  $s$  direction yields:

$$\Phi = A J_n(pr) \cos(n\theta) e^{i(ks - \omega t)}, \quad (20)$$

$$\Psi_s = B J_n(qr) \sin(n\theta) e^{i(ks - \omega t)}, \quad (21)$$

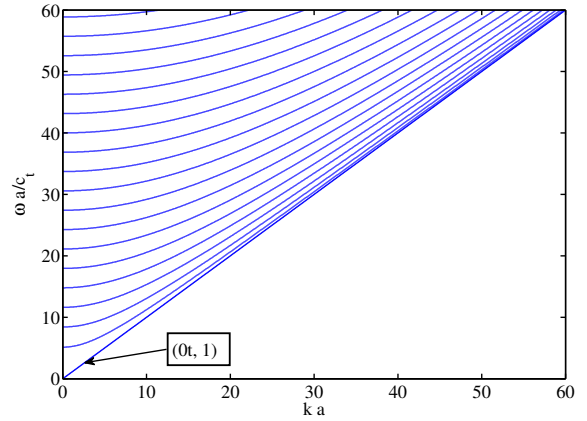
$$\Psi_r = C J_{n+1}(qr) \sin(n\theta) e^{i(ks - \omega t)}, \quad (22)$$

$$\Psi_\theta = -C J_{n+1}(qr) \cos(n\theta) e^{i(ks - \omega t)}, \quad (23)$$

where  $z \mapsto J_n(z)$  is the Bessel function of the first kind and integer order  $n$ ,  $\omega$  is the angular frequency,  $p = (\omega^2/c_l^2 - k^2)^{1/2}$ ,  $q = (\omega^2/c_t^2 - k^2)^{1/2}$ , and  $i = \sqrt{-1}$ . Applying the traction free boundary condition  $\sigma(\mathbf{i}_r)|_{r=a=0}$ , with  $\mathbf{i}_r$  the outward normal of the cross-section, leads to the matrix formulation:

$$\mathbf{\Gamma}_n(s, \theta, \omega, k) \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \mathbf{0}. \quad (24)$$

The matrix  $\mathbf{\Gamma}_n$  directly obtained from the boundary conditions is too cumbersome to be displayed here, but can be found in [20] page 133, Eq. (68). This system has a set of non trivial solutions



**FIGURE 4.** DISPERSION CURVES OF THE FIRST TO THE 36-TH TORSIONAL MODES FOR A CYLINDRICAL BEAM.

if  $\det \mathbf{\Gamma}_n = 0$ . This gives the dispersion relation, that is the relation between the circulation frequency  $\omega$  and the wavenumber  $k$ .

### Analysis of the Frequency Spectrum

Typically, the analysis of the dispersion relation is performed for a fixed value of  $n$ . For  $n = 0$  the  $\theta$ -dependent and  $s, r$ -dependent motions are uncoupled. The first set of motions corresponds to torsional modes. Dispersion curves are obtained by the resolution of:

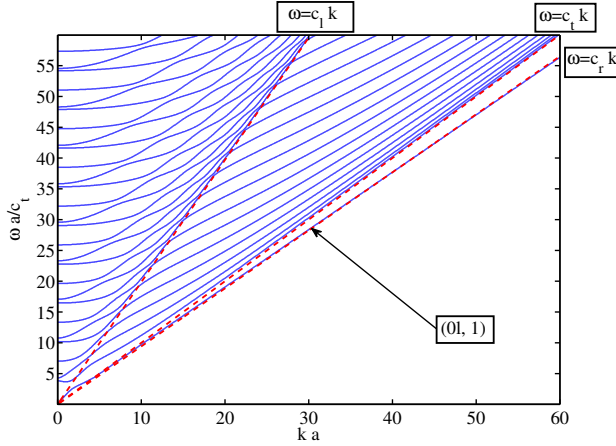
$$qa J_0(qa) = 2 J_1(qa).$$

They are displayed on FIG. 4 as dimensionless circular frequencies  $\omega a/c_t$  as functions of the dimensionless wavenumber  $ka$ . The non-dispersive mode denoted by  $(0t, 1)$  corresponds to the torsional Timoshenko mode because the displacement has the shape  $\mathbf{u}_\theta = B r e^{i(ks - \omega t)}$ , and its velocity is the shear velocity,  $c_{0t,1} = c_t$ . The group velocities of all other torsional modes converge to that shear velocity and their displacements are  $\mathbf{u}_\theta = B J_1(\beta r) e^{i(ks - \omega t)}$ .

The second set of 0-th order modes is constituted by the so-called longitudinal modes. Their dispersion relations come from the resolution of the Pochhammer equation, which is:

$$\begin{aligned} (q^2 - k^2)^2 J_0(pa) J_1(qa) + 4k^2 p q J_1(pa) J_0(qa) \\ = \frac{2p}{a} (q^2 + k^2) J_1(pa) J_1(qa). \end{aligned}$$

A complete survey of this set is given in [21]. FIG. 5 displays the dispersion curves for the first twenty longitudinal modes. As

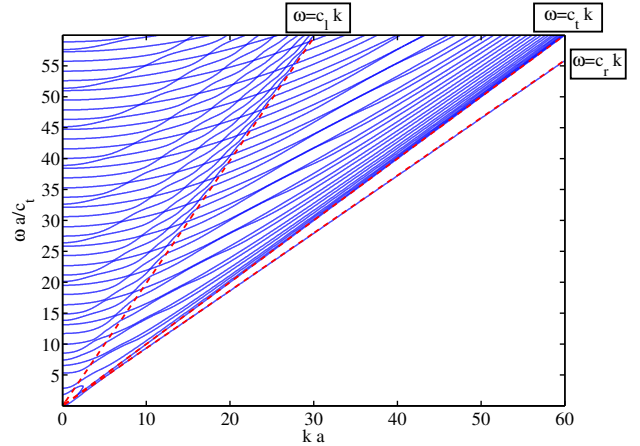


**FIGURE 5.** DISPERSION CURVES OF THE FIRST TO THE 20-TH LONGITUDINAL MODES FOR A CYLINDRICAL BEAM. THE DOTTED RED LINES SHOW THE DISPERSION RELATIONS FOR LONGITUDINAL, TRANSVERSE AND RAYLEIGH WAVES IN A REFERENCE THREE-DIMENSIONAL HOMOGENEOUS AND ISOTROPIC ELASTIC MEDIUM (WITH A POISSON RATIO  $\nu = 0.33$ ).

seen of this plot, all the modes become non-dispersive in the high frequency limit. In addition, It can be shown analytically that in the lower frequency range  $ka \rightarrow 0$ , the group velocity of the mode denoted by  $(0l, 1)$  on FIG. 5 approaches the velocity of the longitudinal mode of the Timoshenko theory (that is  $c_p$ ). The corresponding displacement is constant over the cross-section and purely axial. In the HF range  $ka \rightarrow 0$ , that group velocity now approaches the Rayleigh wave velocity. The displacement is axial and localized close to the surface of the cross-section, similarly to Rayleigh waves in an elastic half-space. All other modes have a similar behavior: at higher frequency, their velocities tend to the shear velocity  $c_t$  and their displacements are mainly radial and antisymmetric. Another interesting property of these curves is that the line  $\omega = c_l k$  is an envelope for all branches. At the intersections between this line and the dispersion curves the cross-section motions are similar to that of the longitudinal Timoshenko mode.

The cases  $n > 0$  correspond to the flexural modes. In the HF frequency range, the velocity of the first mode tends to the Rayleigh velocity while that of the other modes tends to the shear velocity. FIG. 6 displays the first sixteen dispersion curves for  $n = 1$ , the so-called first flexural modes.

The above analyses highlight the fact that Timoshenko theory may be no longer relevant in the very high frequency range, and should probably be adapted and refined in this very limit by considering appropriate group velocities.



**FIGURE 6.** DISPERSION CURVES OF THE FIRST TO THE 16-TH FLEXURAL MODES FOR A CYLINDRICAL BEAM. THE DOTTED RED LINES ARE THE SAME AS FOR FIG. 5.

## CONCLUSIONS

A transient transport model for the propagation of the HF vibrational energy density in three-dimensional beam trusses has been derived. It accounts for the power flow reflection and transmission phenomena at the junctions in this range. The proposed analytical model is integrable by a Runge-Kutta discontinuous Galerkin (RK-DG) finite element scheme. Owing to its low numerical dispersion and dissipation errors, the RK-DG method is suitable for long time simulations. Numerical results obtained for an example of a beam truss illustrate the transient regime and the onset of a diffusive regime at late times. The latter is prompted by the multiple reflections and transmissions of the energy modes at the beam junctions, with possible mode conversions. This scattering process is different from the multiple scattering of waves on random heterogeneities or inclusions considered elsewhere (see [3, 6, 8]), but it gives rise to a similar qualitative and quantitative behavior at late times. Diffusion is characterized by an equilibration of the total energy in each subsystem (a beam of the truss) and for each propagation mode, a situation comparable to the assumption of modal equipartition invoked in the Statistical Energy Analysis (SEA) of structural-acoustic systems. It has been shown finally that the accuracy of the kinematics chosen could be improved in the HF range: this issue is the subject of ongoing researches.

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